

Solution Requirements

Date	5 July 2026
Team ID	SWTID-2026-2638
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should do).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Users can securely register, log in, log out, and verify their identity using email and password.</i>	High
2	Authorization levels	<i>Define user roles (Admin, User, Analyst) and assign access permissions to features and data based on each role.</i>	High
3	External interfaces	<i>Integrate Google Gemini AI API for financial analysis, settlement prediction, negotiation strategies, and letter generation.</i>	High

4	Transactions processing	System manages loan data, financial calculations, settlement processing, payments tracking, and AI-driven recommendations securely.	High
5	Reporting	Generate financial health reports, debt summary reports, settlement analysis, dashboards, and exportable reports for users.	High
6	Business rules, etc.	Implement financial rules such as debt-to-income ratio calculation, settlement eligibility checks, risk assessment, and recommendation logic.	High
7	Compliance to laws or regulations	Ensure compliance with financial regulations, data privacy laws, and industry standards such as GDPR (If applicable) and RBI guidelines.	High
8	Other	AI negotiation letter generation, chatbot assistance, activity logs, audit trail, email notifications, and secure data backup.	High

Step 2: Non-Functional Requirements (NFR)

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S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	<i>The system should provide fast response times for all user actions and AI processing.</i>	<i>e.g., Page loads in < 2 seconds, API response time < 500 ms</i>
2	Scalability	<i>The system should handle growth in users, loans, and transactions without performance degradation.</i>	<i>e.g., Supports up to 10,000 concurrent users</i>
3	Security & Data Privacy	<i>Ensure data protection, encryption, and secure handling of sensitive financial information.</i>	<i>e.g., End-to-end AES-256 encryption, secure authentication</i>
4	Reliability & Availability	<i>The system should be highly available with minimal downtime and proper fault tolerance.</i>	<i>e.g., 99.9% uptime per month</i>
5	Usability & Accessibility	<i>Provide an intuitive user experience and follow accessibility standards for all users.</i>	<i>e.g., Complies with WCAG 2.1 AA standards</i>
6	Other	<i>Ensure maintainability, portability, and easy integration with future features and third-party services.</i>	<i>e.g., Maintainable codebase, cross-browser support, easy deployment</i>